

EML 4507: Finite Element Analysis and Design

Homework 5 - Project

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Design Objective

The design objective for this project is to design a lightweight frame structure capable of supporting a distributed load of 500 N m^{-1} over a 10 m span. The frame structure must not deflect more than 2 mm and must be lighter than the smallest single beam required to achieve this same deformation. The structure must be made of alloy steel and not exceed its yield strength of 620 MPa with a safety factor of 1.5.

Design Method

Description

The frame structure utilizes a Warren Truss configuration to maximize the strength-to-weight ratio. Various combinations of beam cross-sections and truss heights were tested to minimize the weight of the structure. The structure is designed using square bar stock members welded together. A vertical height of 2500 mm was chosen. This height helps increase the overall stiffness, allowing for smaller cross-section members to be used.

Baseline Weight Determination

First, the cross-section for the single beam was calculated. The moment of inertia of the beam can be calculated using the material properties and the maximum desired deflection.

$$\delta_{max} = \frac{5fL^4}{384EI}$$

Since the bar has a square cross-section, the moment of inertia can be related to the side length by the equation:

$$I = \frac{a^4}{12}$$

Plugging in the known values:

$$f = 500 \text{ N m}^{-1}, \quad L = 10 \text{ m}, \quad E = 2.1 \times 10^{11} \text{ Pa}, \quad \delta_{max} = 0.002 \text{ m}$$

Solving gives:

$$I = 1.55 \times 10^{-4} \text{ m}^4, \quad a = 0.207 \text{ m}$$

Finally, the weight of the beam can be solved for using the following equations:

$$W = mg, \quad m = \rho V, \quad V = La^2$$

$$W = \rho La^2 g$$

Given:

$$\rho = 7700 \text{ kg/m}^3$$

Solved:

$$W = 32\,333.86 \text{ N}$$

This beam was tested in SOLIDWORKS to confirm that it meets the requirements. The beam deflected exactly 2 mm and exhibited a maximum stress of 4.23 MPa, which is well below the maximum stress allowed.

Concepts

Choosing a taller frame increases the moment of inertia of the entire structure, which allows for less mass to be used. A single beam was used as the base chord to provide a strong foundation for the structure. This beam had the largest cross-section of 40 mm × 40 mm since this is where the distributed load was applied. The top chord and diagonal members all used 10 mm × 10 mm beams to reduce weight. The triangular structure created by the diagonal beams helped transfer the vertical distributed load into axial tension and compression, minimizing bending moments.

Results

Model

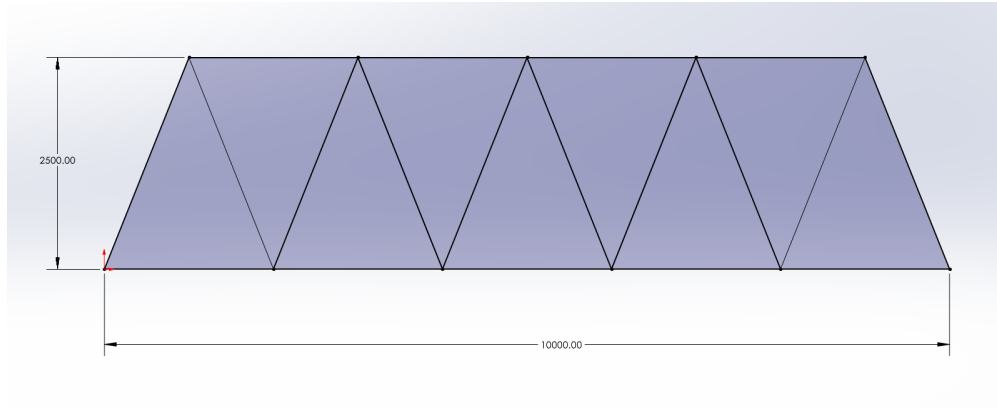


Figure 1: Designed Frame Structure

Stress Criterion

The maximum stress in the frame must be less than the yield strength of the material with a safety factor of 1.5. This means the following equation must be true:

$$\sigma_{max} < \frac{\sigma_y}{1.5} = \frac{620 \text{ MPa}}{1.5} = 413.33 \text{ MPa}$$

Stress Plot

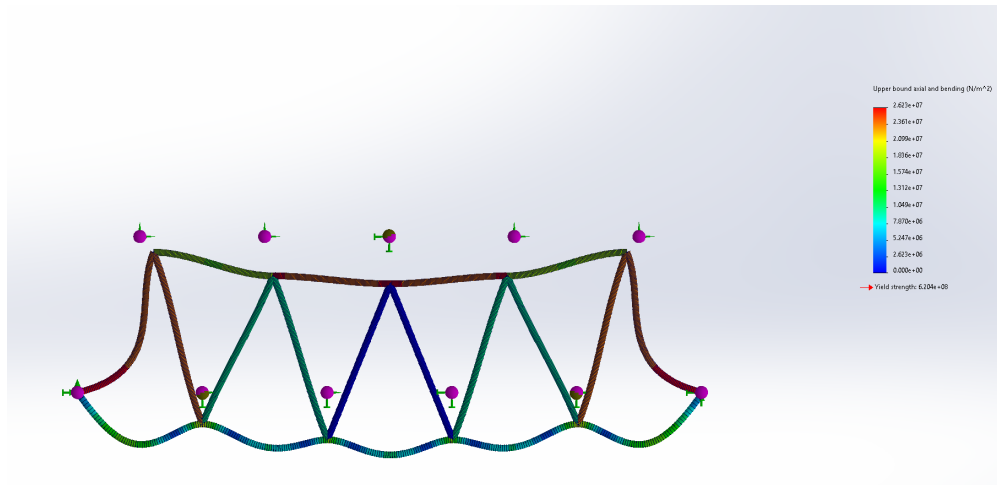


Figure 2: Stress Plot

As shown in Fig. 2, the maximum stress in the structure is 26.23 MPa, which is well below the safety factor requirement.

Displacement Criterion

The maximum displacement in the structure must not exceed 2 mm.

Displacement Plot

As shown in Fig. 3, the maximum displacement in the structure is 1.914 mm, which is below the maximum allowed.

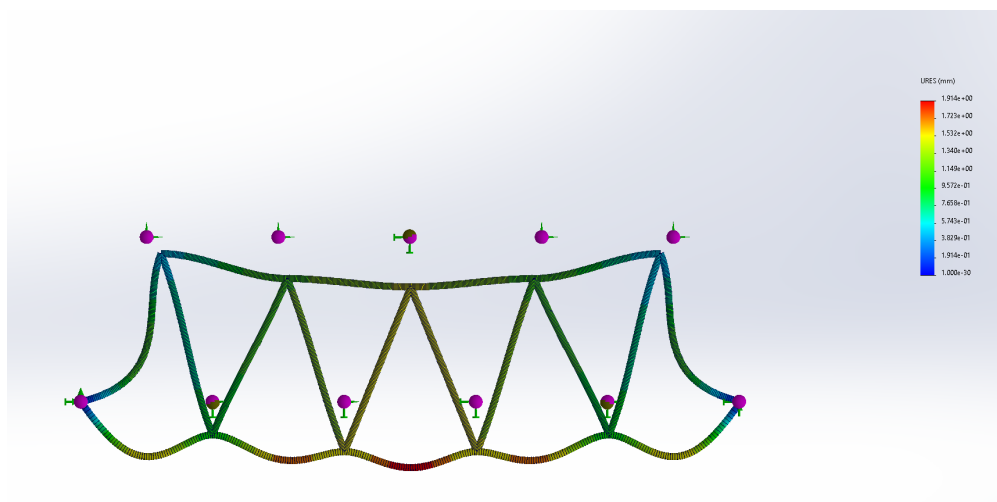


Figure 3: Displacement Plot

Member Weight Table

Table 1: Dimensions and Mass of Truss Members

Member ID	Length (mm)	Side Length (mm)	Mass (kg)
1 (Bottom Chord)	10,000	40	123.20
2 (Top Chord)	2000	10	1.54
3 (Top Chord)	2000	10	1.54
4 (Top Chord)	2000	10	1.54
5 (Diagonal)	2692.58	10	2.073
6 (Diagonal)	2692.58	10	2.073
7 (Diagonal)	2692.58	10	2.073
8 (Diagonal)	2692.58	10	2.073
9 (Diagonal)	2692.58	10	2.073
10 (Diagonal)	2692.58	10	2.073
11 (Diagonal)	2692.58	10	2.073
12 (Diagonal)	2692.58	10	2.073
13 (Diagonal)	2692.58	10	2.073
14 (Diagonal)	2692.58	10	2.073
15 (Diagonal)	2692.58	10	2.073
Total	—	—	150.62

Conclusion

The final weight of the structure is 150.62 kg (1477.58 N), which is 95.43 % lighter than the baseline single beam. This demonstrates the efficiency of truss structures in reducing weight for engineering applications. After testing with different numbers of diagonal elements, it was clear that adding diagonal elements increases the overall stiffness, which allows for lighter individual beams to be used. This trade-off between design complexity and weight reduction is a critical tool for structural design.